Solution Requirements

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Farmers, Researchers, and Admin users securely log in using username and password	High
2	Authorization levels	Admin manages datasets and users; Farmers receive recommendations; Researchers can analyze crop data.	High
3	External interfaces	Integrates weather API, rainfall data, and crop dataset for prediction.	High

4	Transactions processing	Processes soil values (N, P, K, pH), temperature, humidity, and rainfall to generate crop recommendations.	High
5	Reporting	Displays crop prediction results, environmental suitability, and recommendation history.	Medium
6	Business rules, etc.	Recommend the most suitable crop based on environmental conditions and trained ML model.	High
7	Compliance to laws or regulations	Protect user information and follow basic data privacy and security practices.	Medium
8	*Other*	Validate all input values before prediction and prevent invalid entries.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate prediction quickly after user input.	Response time < 2 seconds

2	Scalability	Support increasing users and larger agricultural datasets.	1000+ concurrent users
3	Security & Data Privacy	Secure authentication and encrypted password storage.	Secure login & protected database
4	Reliability & Availability	System should remain operational with minimal downtime.	99% uptime
5	Usability & Accessibility	Simple interface usable by farmers with minimal technical knowledge.	Responsive UI for desktop & mobile
6	*Other*	Modular codebase for easy maintenance and future enhancements.	Well-documented and maintainable